

Module 1: IPv6 Addressing

Topic 1.1: Intro to IPv6

Intro to IPv6

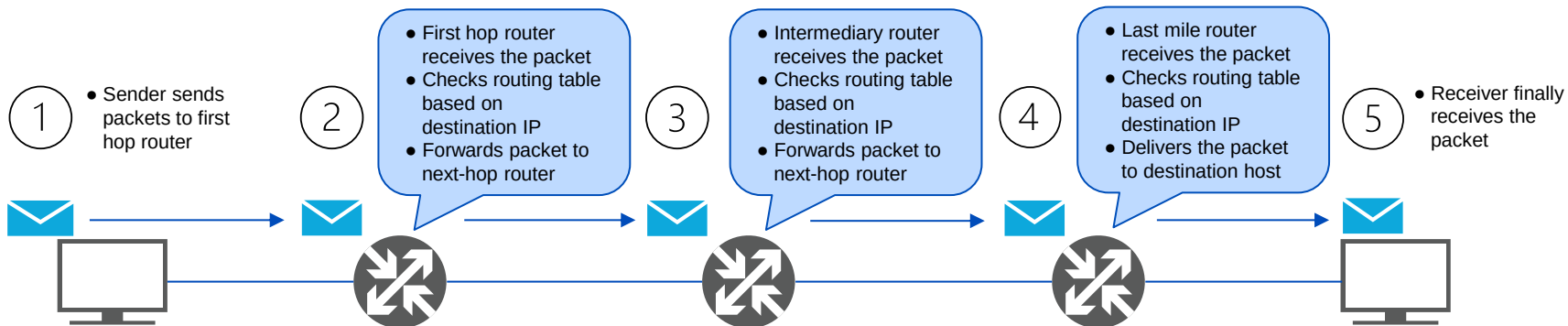
- Exam Objectives:
 - What is IPv6?
 - The reason behind IPv6?

Topic 1.1: Intro to IPv6

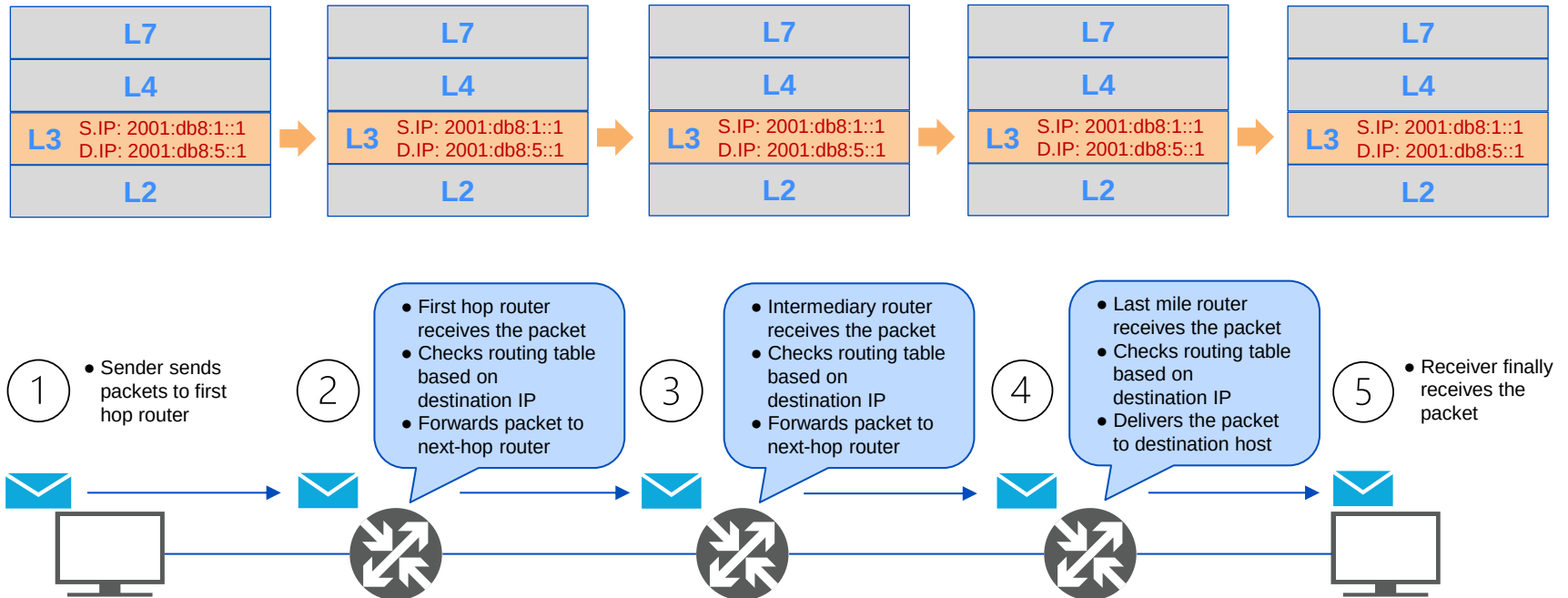
What is IPv6?

What is IPv6?

- IPv6 is a variant of IP protocol (like IPv4)
 - Helps to **identify a host uniquely** on the internet
 - Helps to **carry packets** across the network
 - Can be used in **source and destination address** fields of an IP packet
- IPv6 is a **next-gen** protocol



What is IPv6?



What is IPv6?

- Much bigger address space
 - To address IPv4 exhaustion problem and so on

IPv4: A **32-bit** long address space



11000000 . 10101000 . 00001010 . 00000001

IPv6: A **128-bit** long address space



0010 0000 0000 0001 : 0000 1101 1011 1000 : 0000 0000 0000 0000 : 0000 0000 0000 0000 : 0000 0000 0000 0000 : 0000 0011 0110 1110 : 0001 0010 0101 0000 : 0010 1011 0000 0000

What is IPv6?

- How much bigger the IPv6 is?
 - Is it **4 times** bigger than IPv4?
 - 33 bit is double than 32 bit
 - 34 bit is double than 33 bit
 - 35 bit is double than 34 bit
 - ⋮
 - 128 bit is double than 127 bit

IPv6 = IPv4 x 10²⁹



IPv6: 340282366920938463463374607431768211456

Can we pronounce this?

340 undecillion

282 decillion

366 nonillion

920 octillion

938 septillion

463 sextillion

463 quintillion

374 quadrillion

607 trillion

431 billion

768 million

211 thousand

456



Topic 1.1: Intro to IPv6

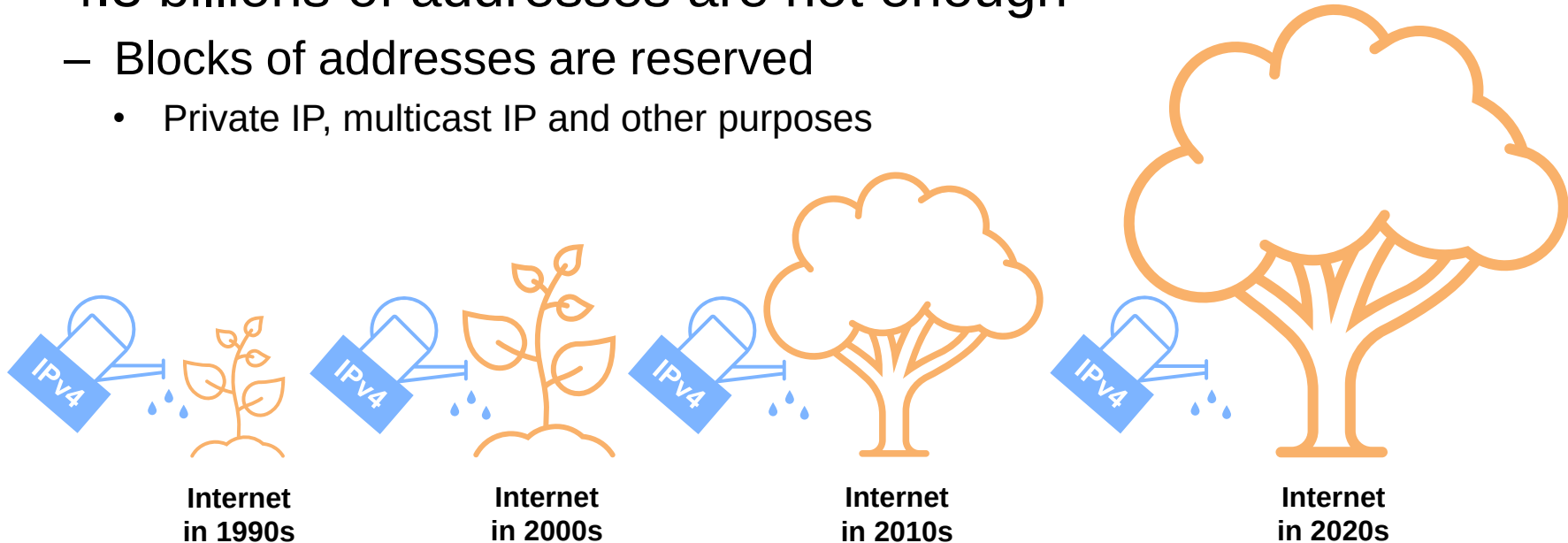
The reason behind IPv6?

The reason behind IPv6?

- IPv4 address space exhaustion
- Problems with NAT in IPv4
- Implementation of future IoT
- Simplified header for transmission
- Moving forward to SDN era

IPv4 address space exhaustion

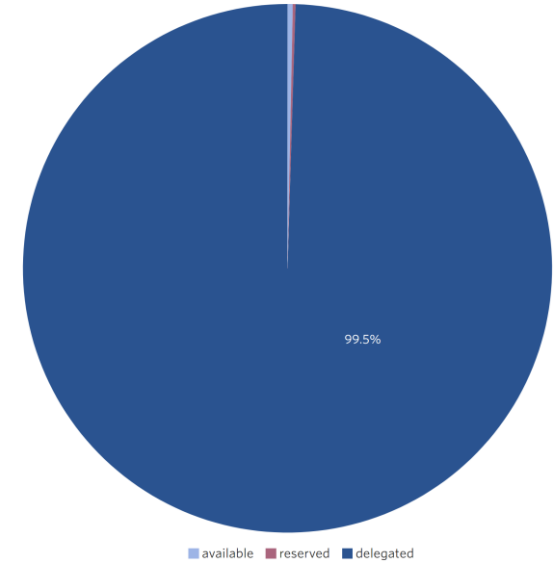
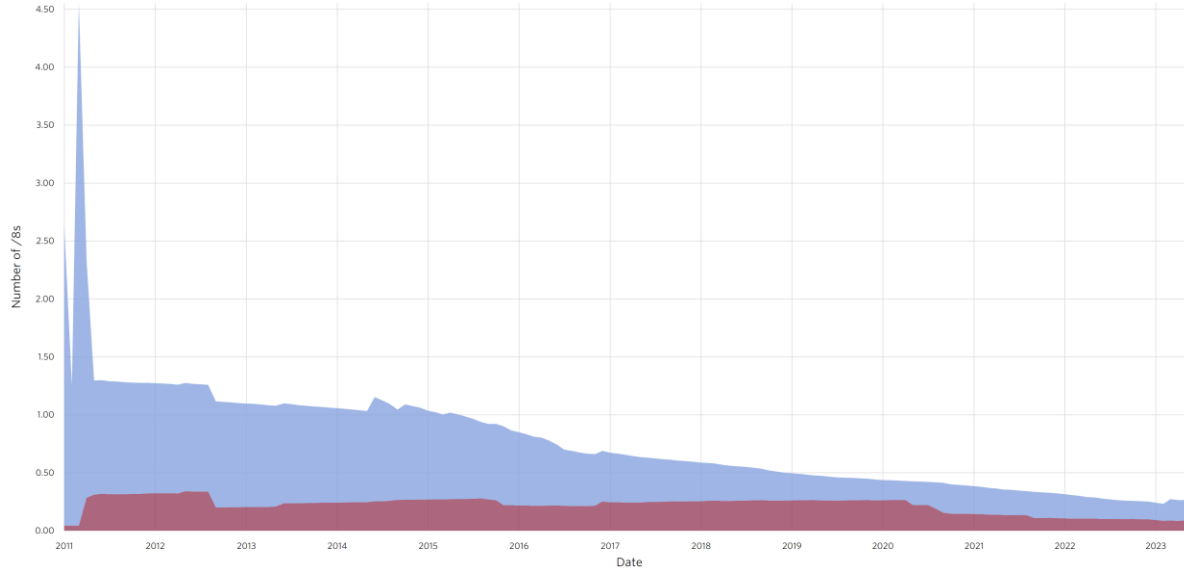
- Internet is growing rapidly
- 4.3 billions of addresses are not enough
 - Blocks of addresses are reserved
 - Private IP, multicast IP and other purposes



IPv4 address space exhaustion

- Inefficient allocation of addresses from the very beginning
 - Early allocation model contributed to waste of addresses
 - Large: up to 16,777,216 hosts (Class A)
 - Medium: up to 65,536 hosts (Class B)
 - Small: up to 256 hosts (Class C)
 - CIDR has been introduced in 1993
 - NAT has been introduced in 1994
 - RFC1918
- APNIC received the last two blocks of IPv4 On January 31, 2011

IPv4 address space exhaustion



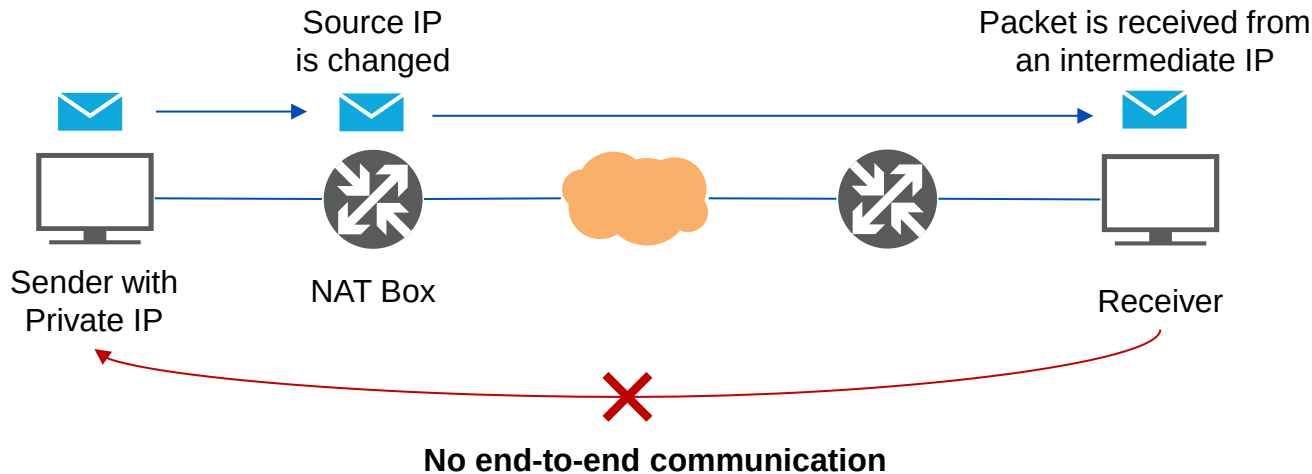
APNIC's IPv4 Free Pool Chart

IPv4 address space exhaustion

- Worth reading:
 - <https://www.apnic.net/manage-ip/ipv4-exhaustion>
 - <https://www.ripe.net/manage-ips-and-asns/ipv4/ipv4-run-out>
 - <https://www.apnic.net/community/ipv6>

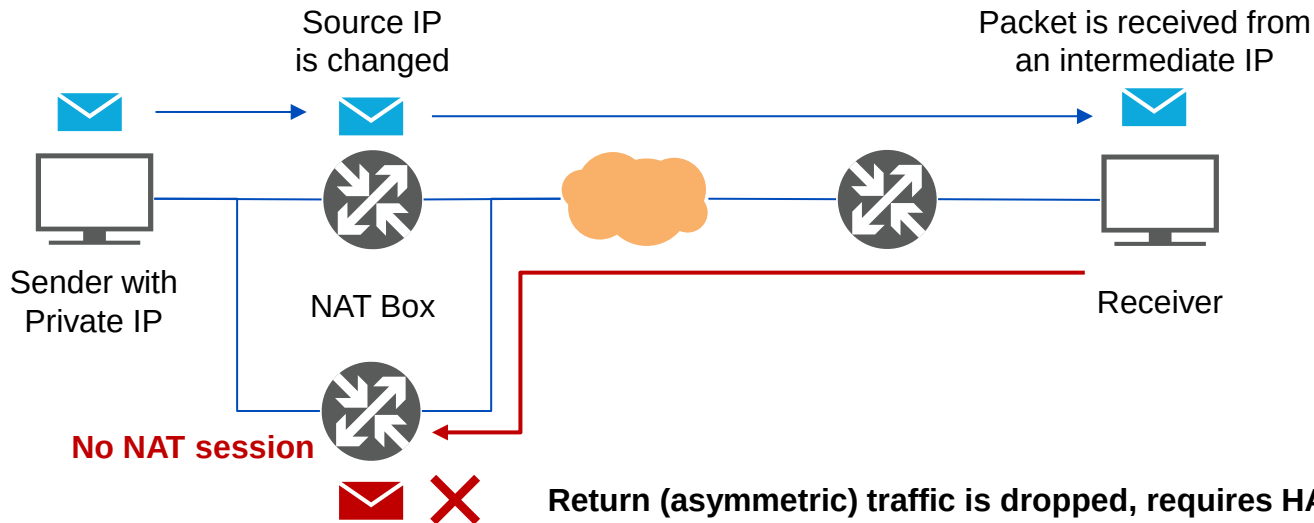
Problems with NAT in IPv4

- Breaks end-to-end connectivity
 - Destroys peer-to-peer networking model
 - Not a true contributor to security enforcement



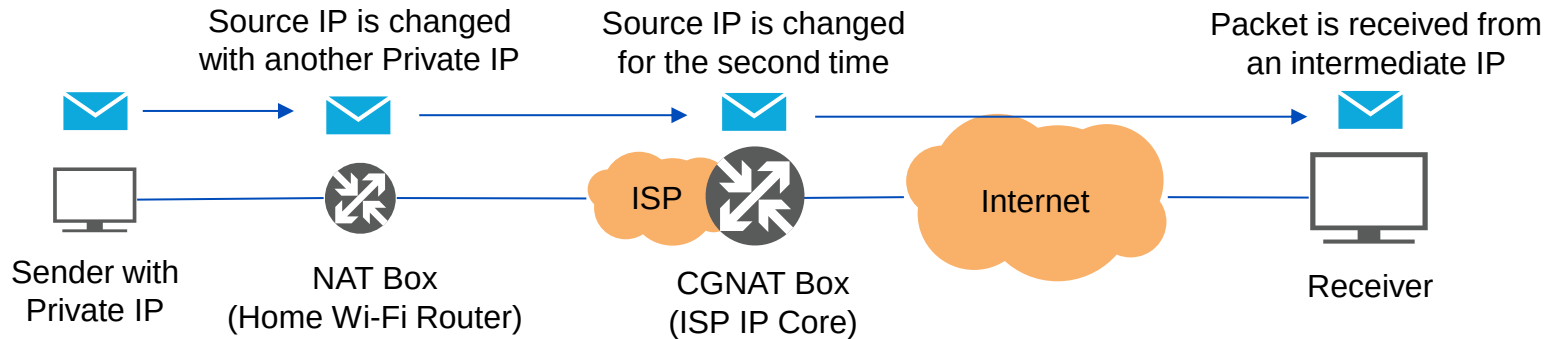
Problems with NAT in IPv4

- Breaks end-to-end connectivity
 - Breaks communication while traffic path is asymmetric
 - Requires HA clusters configured; introduces higher costs



Problems with NAT in IPv4

- Requires CGNAT devices for service providers
 - Adds extra processing on NAT box
 - Adds extra costs and efforts to deploy
 - Actual source addresses are hard to be identified for LEA



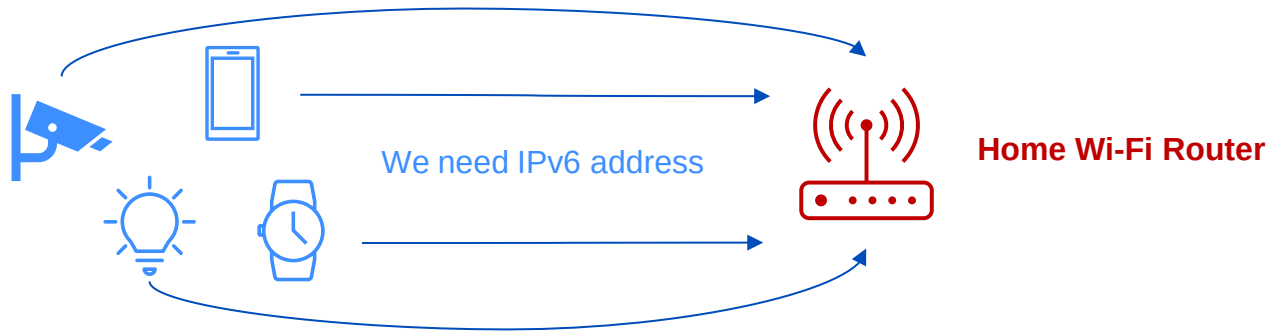
Implementation of future IoT

- IPv6 to support a huge number of IoT devices
 - IPv6 helps to provide end-to-end connectivity
 - No NAT is intended
 - True peer-to-peer model



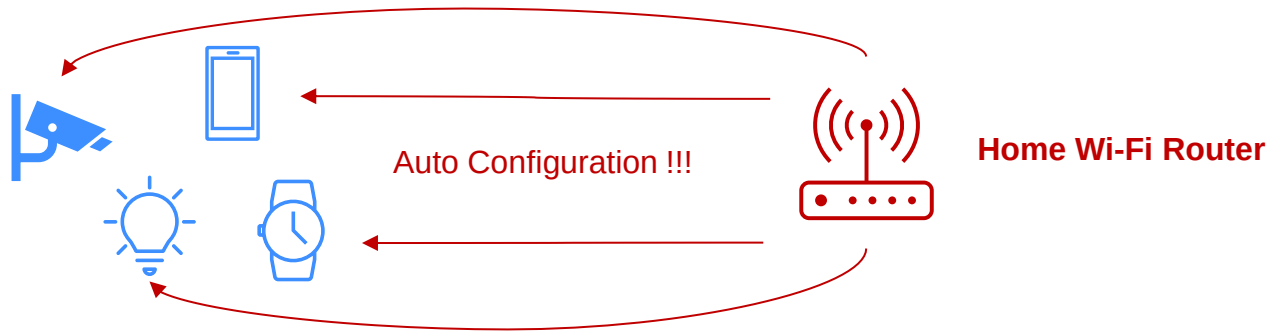
Implementation of future IoT

- IPv6 to support a huge number of IoT devices
 - Default support of address auto configuration
 - DHCP server is not mandatory



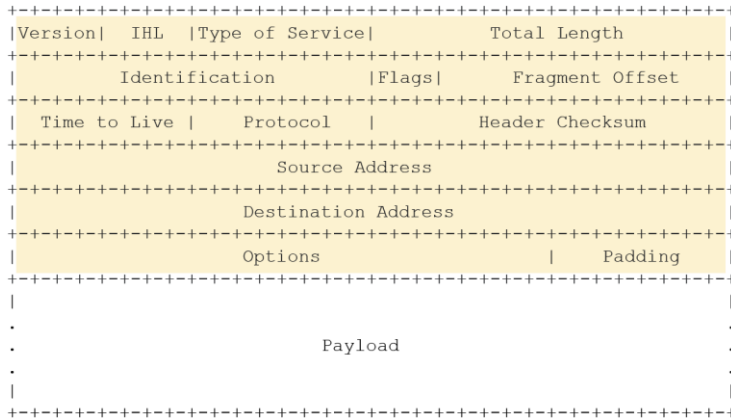
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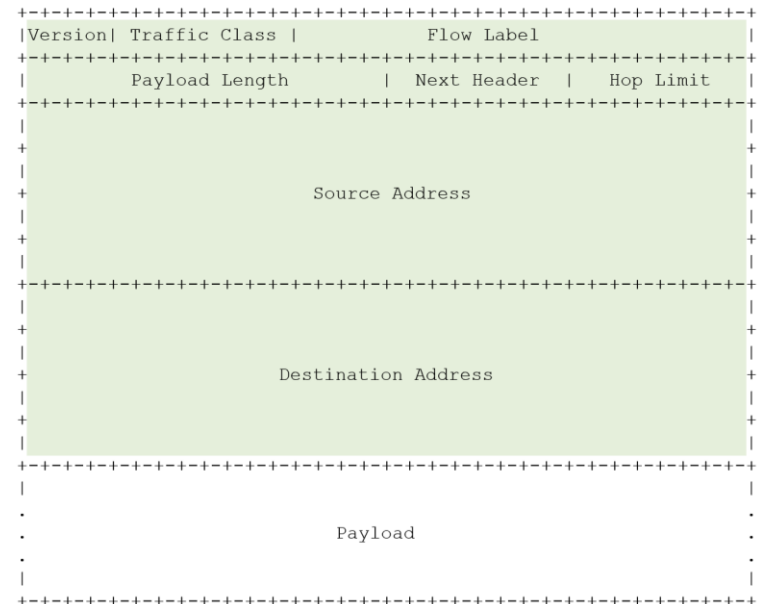


Simplified header for transmission

- IPv6 has less fields in its header
 - Helps in faster packet processing
 - No header checksum check in every hop
 - Fixed length header
 - Scalable for deployment and cost effective



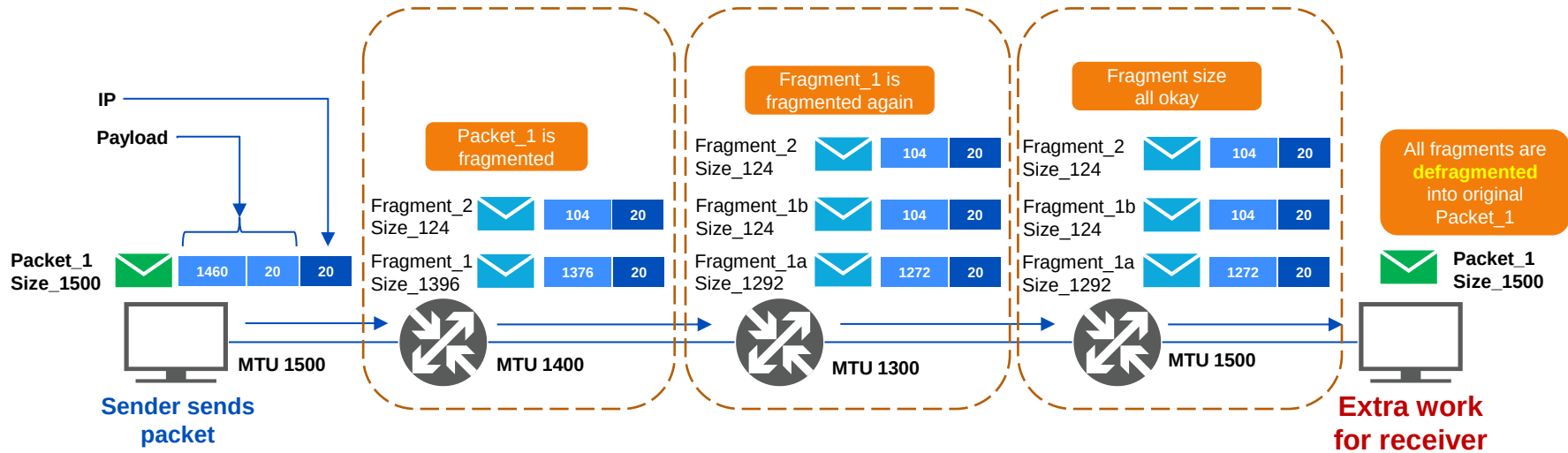
IPv4 Header



IPv6 Header

Simplified header for transmission

- PMTUD in IPv6 reduces extra load on receiving devices

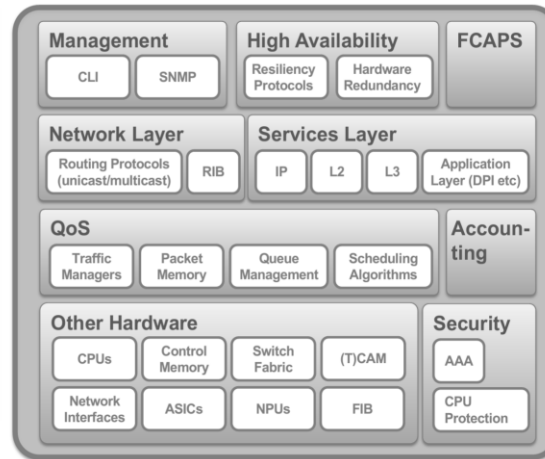


Packet fragmentation in IPv4

Moving forward to SDN era

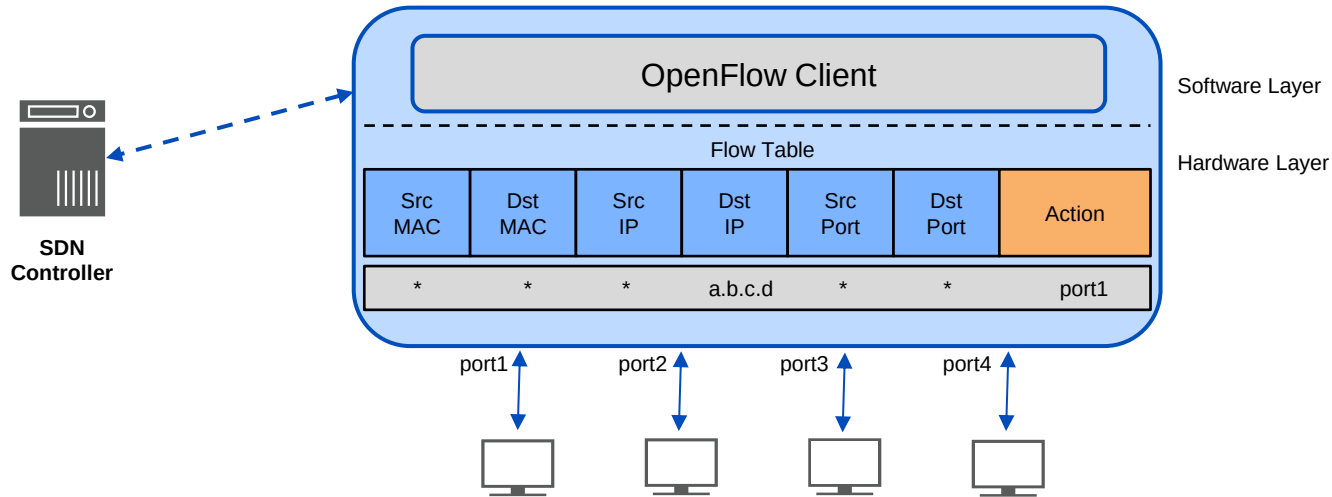
- Current IP network has many limitations
 - Lack of network and traffic visualization
 - Lack of network programmability and automation
 - Control plane and data plane are not decoupled

Router now a days



Moving forward to SDN era

- SDN can solve the problems
 - Prof. Nick McKeown's team proposed the new network architecture



Moving forward to SDN era

- Software Defined Networking (SDN)
 - Control plane and data plane are decoupled
 - Control plane: To deal with control logics
 - Data plane: To execute instructions given by controller
 - Centralized network controller
 - Open network programmability
 - Better automation capability

SDN Technologies

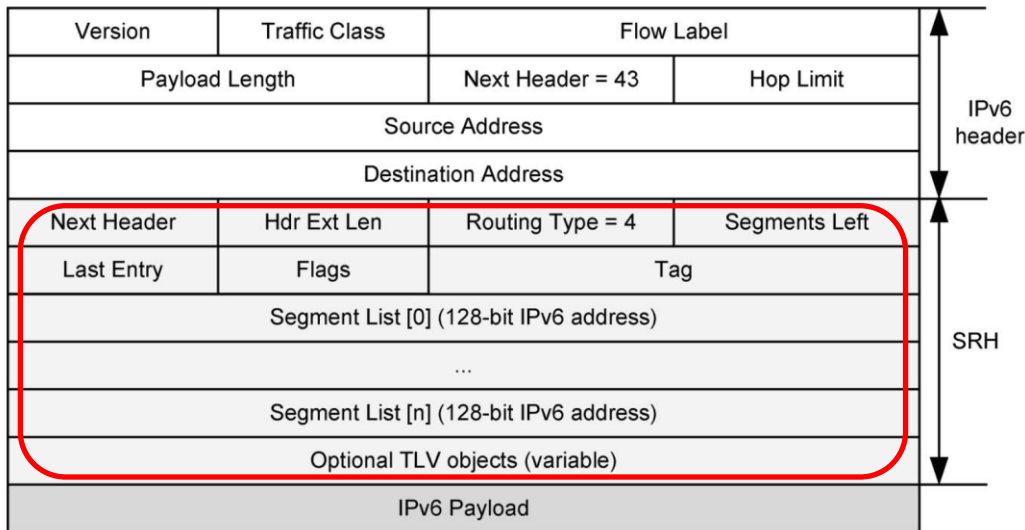
- OpenFlow
- Programming Protocol-independent Packet Processor (P4)
- Segment Routing (SR)
 - SR-MPLS: with MPLS data plane
 - To eliminate the need of LDP and RSVP for MPLS; an IGP extension
 - SRv6: with IPv6 data plane
 - To eliminate the need of MPLS itself
 - Based on native IPv6 forwarding capability; IPv6 extension header

SRv6 for SDN

- A source routing paradigm mostly for carrier IP backbones
- Business benefits is seen in SRv6
 - Doesn't impose strict decoupling of control and data plane
 - Can be used to excel data plane functionality first
 - Fast Re-Route (FRR) and Traffic Engineering (TE)
 - Control plane separation and programmability are the next move
 - BGP-LS, PCEP, NETCONF, YANG
- 5G and Cloud services require network programmability
 - IPv6 Extension Header provides more extensibility

SRv6 Data Plane Overview

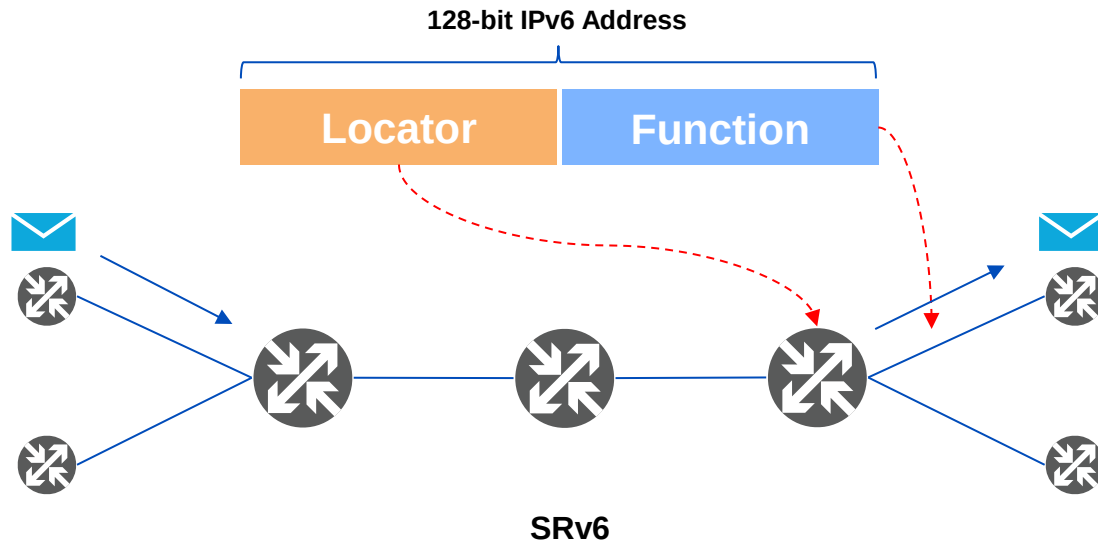
- IPv6 Extension Header: Segment Routing Header (SRH)
 - Protocol Number: 43; Type: 4



Segment Routing Header (SRH)

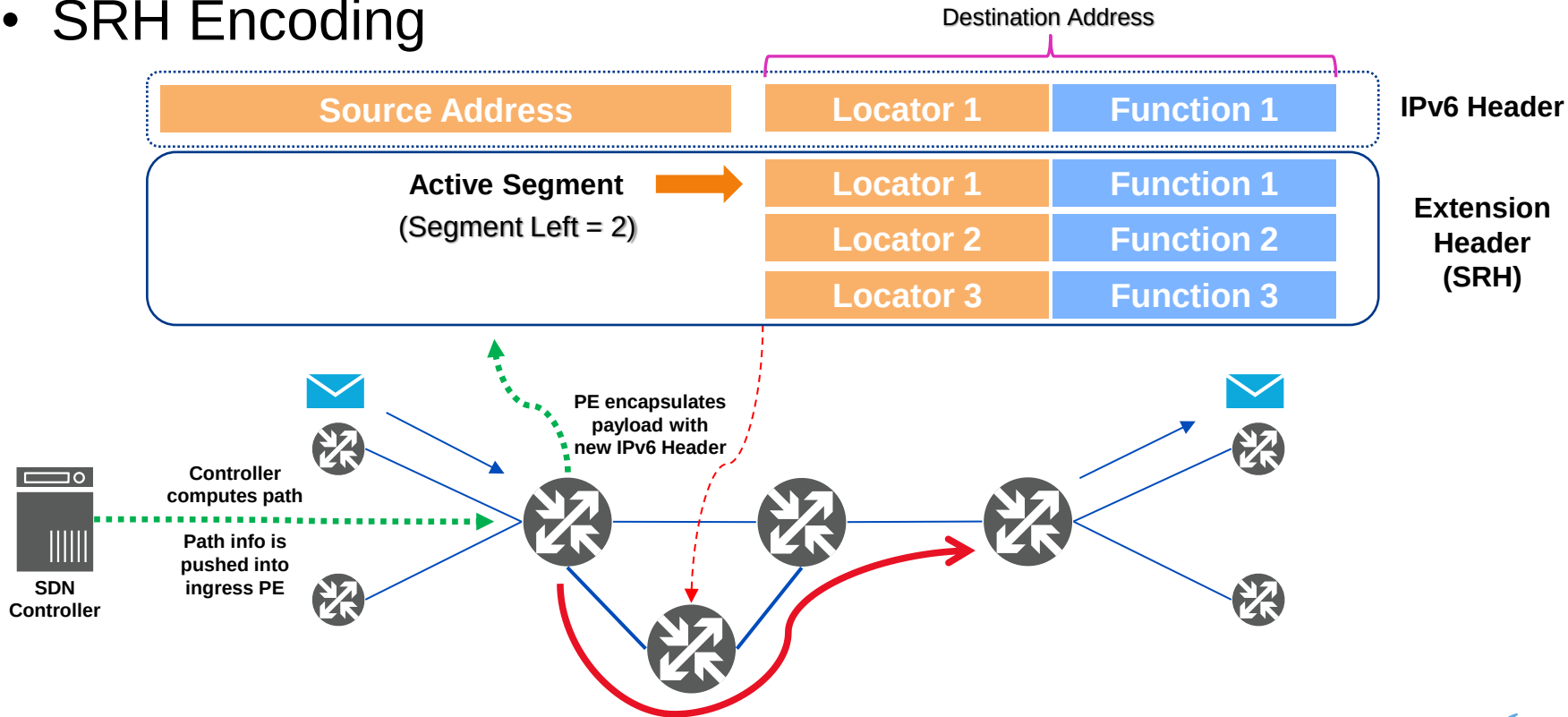
SRv6 Data Plane Overview

- 128-bit SRv6 SID
 - Locator: Defines the node performing the function
 - Function: Defines any possible function (such as service VPN)



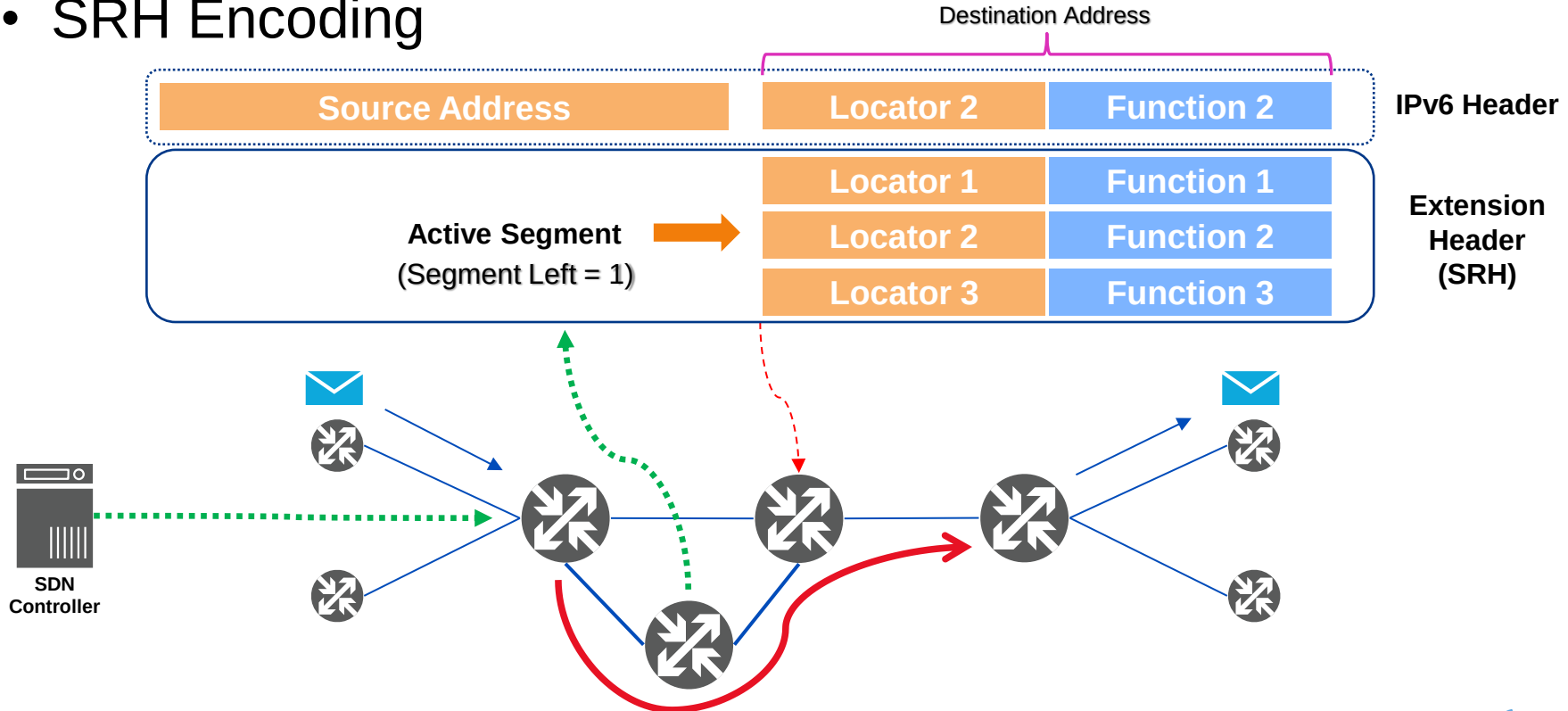
SRv6 Data Plane Overview

- SRH Encoding



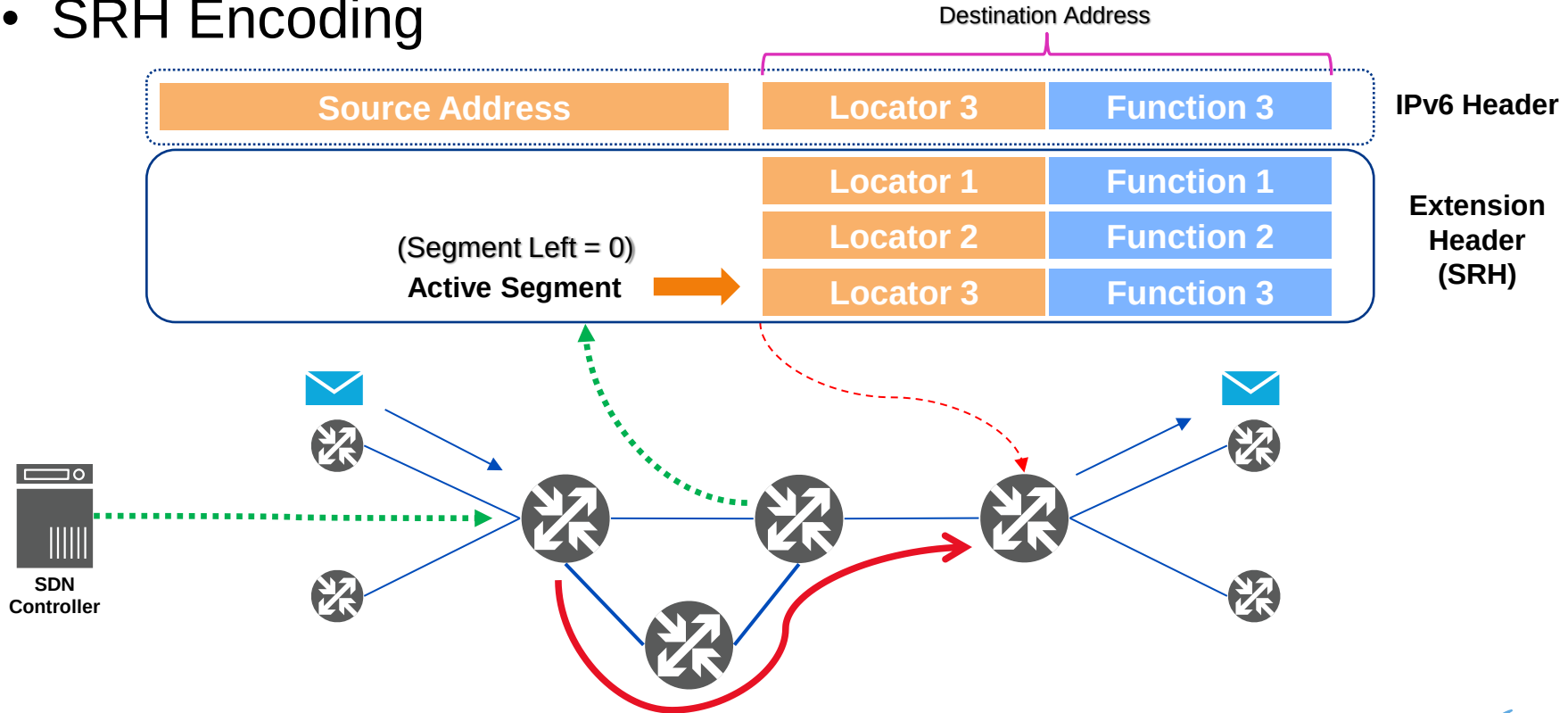
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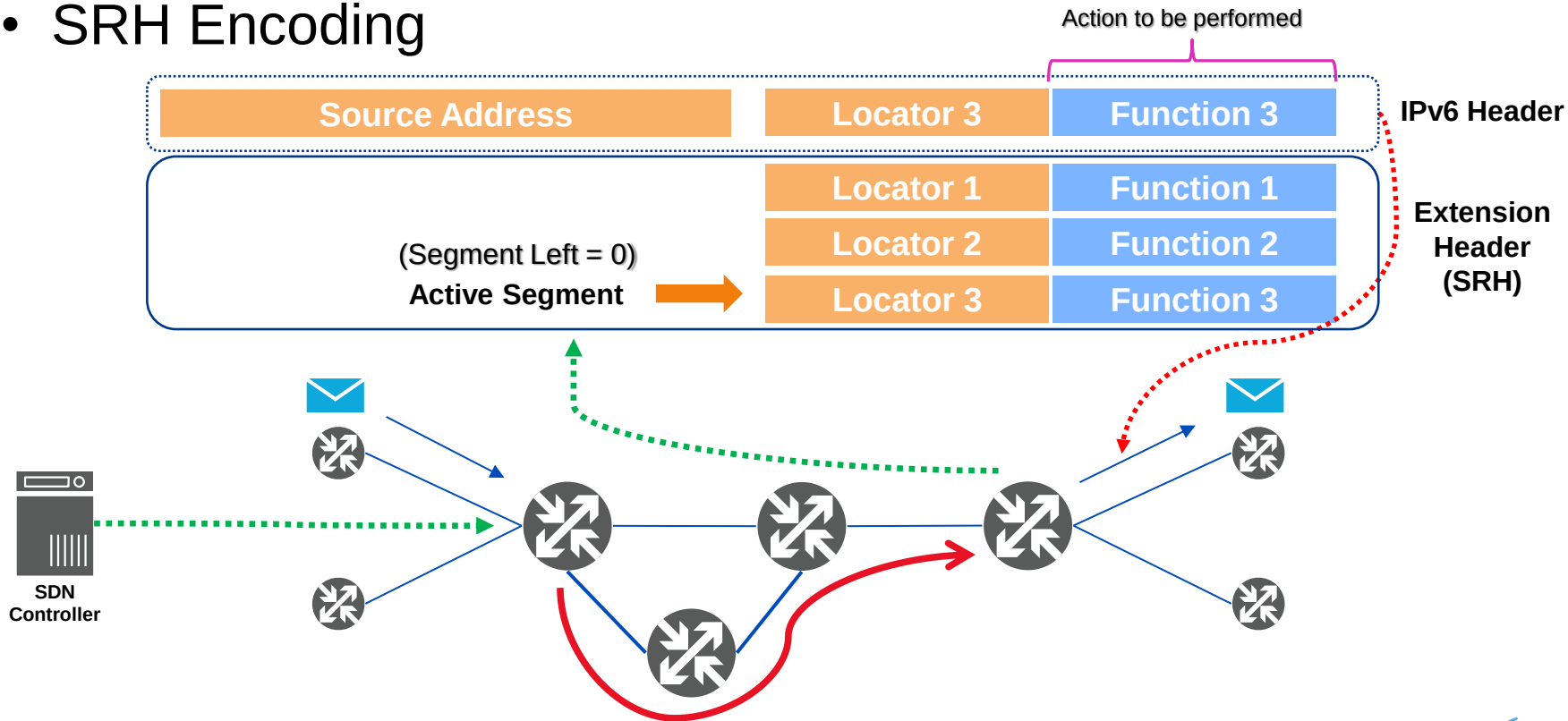
SRv6 Data Plane Overview

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SRv6 Data Plane Overview

- SRH Encoding



Activity



Discussion